

ML Final Project Proposal

Group 03

We are participating in the MALLORN Astronomical Classification Challenge, a binary classification competition on Kaggle. The task involves identifying Tidal Disruption Events (TDEs)—rare phenomena where stars are torn apart by black holes—from simulated LSST astronomical lightcurves. Given the highly imbalanced dataset with very few TDEs among other transients, we need to develop machine learning models that can accurately detect these scientifically valuable events based on brightness variations over time, evaluated using the F1 score.

Dataset

Source and overview

All the dataset is a simulated astronomical dataset based on real nuclear transient observations from ZTF, created for LSST classification tasks. It contains time-series lightcurve data and metadata (e.g., redshift, extinction) for extragalactic transients. The goal is to classify each object as a Tidal Disruption Event (TDE) or not (a binary classification task).

Number of features, size, type

- Features: 3 - Time(MJD), Flux, Flux_err
- Size: 26321
- type : tabular

	object_id	Time (MJD)	Flux	Flux_err	Filter
0	Dornhoth_fervain_onodrim	63314.4662	-1.630159	0.365777	z
1	Dornhoth_fervain_onodrim	63780.9674	10.499389	0.253867	r
2	Dornhoth_fervain_onodrim	63789.7693	5.866250	1.559241	y
3	Dornhoth_fervain_onodrim	63794.1702	3.903623	0.376854	r
4	Dornhoth_fervain_onodrim	63794.1702	5.226644	0.518864	i
...	...	...	...	...	...
26319	yif_gath_dannas	62889.0896	-0.047923	0.165270	i
26320	yif_gath_dannas	62889.0896	-0.353881	0.496303	z
26321	yif_gath_dannas	62893.7636	-0.167742	0.497918	u

Dataset Link

<https://www.kaggle.com/competitions/mallorn-astronomical-classification-challenge/data>

Task Description

Feature Engineering

In order to have more clear, structured data, we will split the data according to the band.

Also, the light sometimes would be obscured by dust on its path to reach us, so we will use the extinction coefficient value to de-extinct the flux measurements with the fitzpatrick99 method.

Model Implementation

We plan to leverage Random Forest as the primary base learner combined with ensemble methods to tackle the severely imbalanced TDE classification task. We will use Random Forest with class weighting to address the minority class problem, then combine their predictions using voting or stacking ensemble techniques.

Evaluation Metric

Submissions will be evaluated using the F1 score, with the primary objective of correctly identifying TDEs within the dataset. Because the dataset is highly imbalanced, with TDEs being significantly rarer than other classes, the F1 score is preferred over simple accuracy in this task. F1 score provides a balanced measure of performance, rewarding models that achieve a good trade-off between recall and precision.